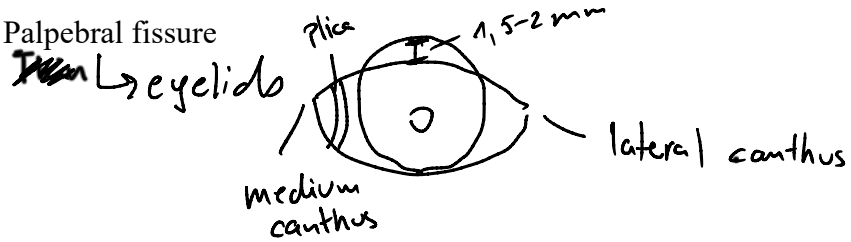
Ocular Adnexa

Images used: 2.2, 2.5, 2.10, 2.1, 2.8, 2.9, 2.12, 2.14

I. External features of the eyelids

A. Palpebral fissure



B. Lateral and medial canthus sides

1. Plica semilunaris - or just plica
raisd tissue
it enlarged when we look lateral

2. Caruncle
middle of it + it is tissue
pink tissue

C. Eyelid Margin

1. Cilia = eyebrows

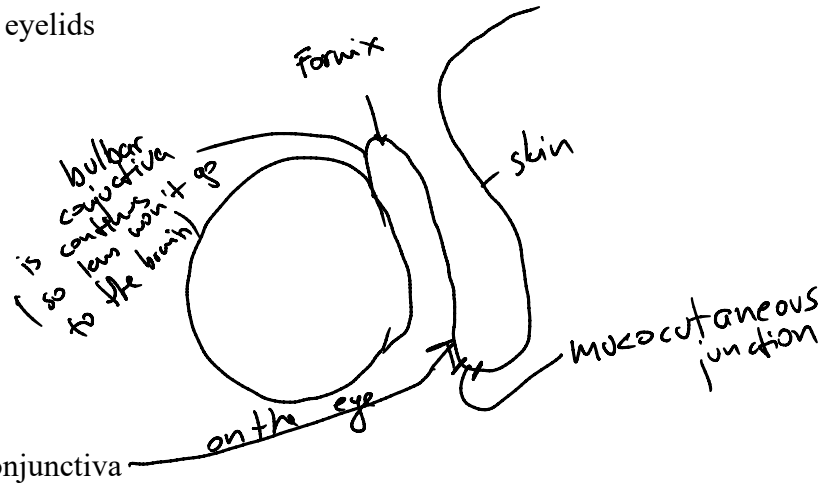
2. Pores of the meibomian glands - outermost tear



- post
- cili portion

II. Anatomy of the eyelids

A. Skin



B. Palpebral conjunctiva

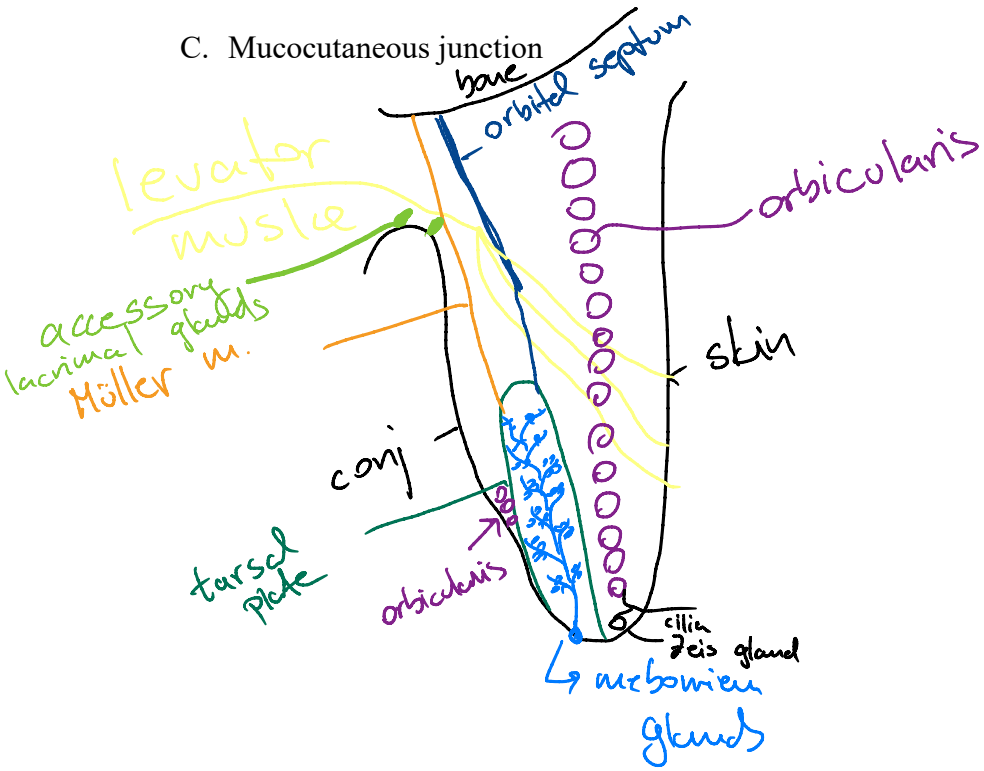
1. Fornix

2. Goblet cells

produce the innermost of the tears

innermost tear layer

C. Mucocutaneous junction



D. Tarsal plate

1. Dense connective tissue that gives the eyelid rigidity and structure

2. The superior tarsal plate is approximately 8 mm high in Asians and 10 mm in Caucasians.
can't bend

3. The inferior tarsal plate is approximately 5 mm high

4. Clinical correlate: Eyelid eversion

E. Orbital septum

1. The orbital septum is a Thick sheet of connective tissue that starts at the orbit rim and inserts into the levator aponeurosis above the tarsal plate superiorly and the tarsal plate inferiorly.

2. Function

   a. Separates the eyelid structures from the orbital contents.

   b. Prevents eyelid infections from entering the orbit and therefore having a direct connection to the brain.
People can die if it comes in!

F. Eyelid muscles

1. Orbicularis oculi muscle in circle
   help us blink.
   closes the eyelid → protects the eye. Have tear film throug the eye.

   a. Encircles the palpebral fissure.

   b. Functions

      (1) Closes the eyelids

      (2) Protects the eye

      (3) Renews the tear film

      (4) Keeps the eyelid margin near the globe

      (5) Helps with meibomian gland secretion

   c. Clinical correlate: ectropion and entropion
      ↳ outward
      - weakes of orbicularis m.
      ↳ inwards (scaring
      - over action of orbicu. m.

   d. Innervation: cranial nerve 7
      ↳ facial nerve

2. Superior palpebral levator muscle spread out through the whole lid

   a. The levator muscle starts in the posterior orbit. →

   b. It travels forward with the superior rectus muscle.

   c. Levator aponeurosis

      (1) Tendinous expansion of the levator - not a m.

   d. Function

      (1) Elevation of the eyelids

      (2) Connections between the superior rectus and the levator results in increased eyelid height when looking up.

   e. Innervation: cranial nerve 3 oculomotor

3. Müller muscle - automatic nerves

   a. Smooth muscle

   b. Inserts into the tarsal plate

   c. Function

      (1) Additional eyelid elevation - similar to levator, but! Müller is controlled by sympatetic n. system.

   d. Innervation: sympathetic nervous system

4. Clinical correlate: Ptosis
    a. A drooping of the upper eyelid.
        → Müller → subtle
        levator → dramatic

G. Glands of the eyelids

1. Meibomian glands
    in tarsal plate
    a. Embedded in the tarsal plate
    b. Function: secrete the outer lipid layer of the tear film
        important in dry eye

2. Zeis glands
    a. Secretes sebum into the hair follicle
    b. Function: keep the eyelash from becoming brittle.

3. Accessory lacrimal glands
    → to do with tears
    a. Located in the upper and lower conjunctival fornix and orbital borders of the tarsal plate
    b. Function: contributes to the aqueous layer of the tear film

III. Lacrimal system

A. Tear film

1. Functions

a. Lubricant between the globe and eyelids

b. Helps remove debris (dirt, ...)

c. Source of oxygen (cornea? no vessels)

d. Antibacterial substances (protect from bacteria)

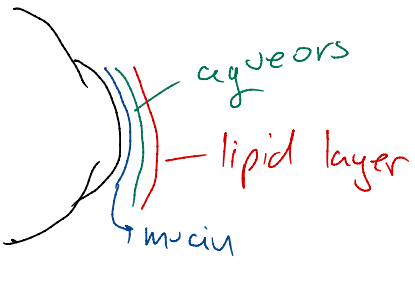
e. Maintains corneal hydration

f. Regulate ocular surface wound repair

2. Layers

a. Lipid layer

(1) Produced in meibomian glands



(2) Function

i. Retards evaporation (ochlapewnie?)

ii. Provides lubrication

iii. Stabilizes the tear film (lowering surface tension) to keep tears from overflowing down the cheeks

b. Aqueous layer watery layer

(1) Secreted by the lacrimal and accessory lacrimal glands → ce mai kuj v acesu ipride il teq

inside the eyes

4 lateral 4 superior

help to produce tears

c. Mucin layer closest to the cornea
tracks the water and sticks it to cornea.

(1) The surface of the conjunctival and corneal cells has microvilli covered with this glycocalyx layer which is hydrophilic.

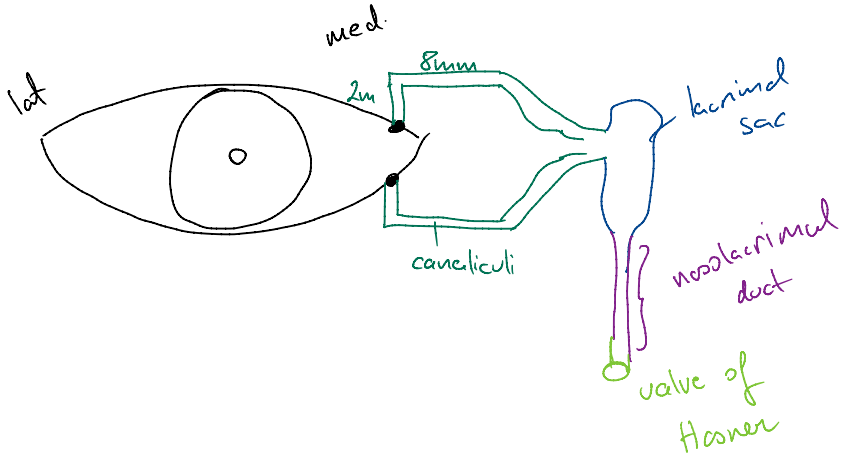
(2) Function

i. Facilitates adhesion of the aqueous layer to the ocular surface

ii. Allows the aqueous layer to spread evenly over the surface of the eye

B. Nasolacrimal drainage system - how the tears are draining

1. Puncta
Where the tears drain upper and down lid



2. Canaliculi

3. Lacrimal sac

4. Nasolacrimal duct

5. Valve of Hasner: found at the end of the nasolacrimal duct.